

Motivation

Channel **dependency** refers to the **necessity of using multiple channels** for successful classification, not just statistical correlation between channels.

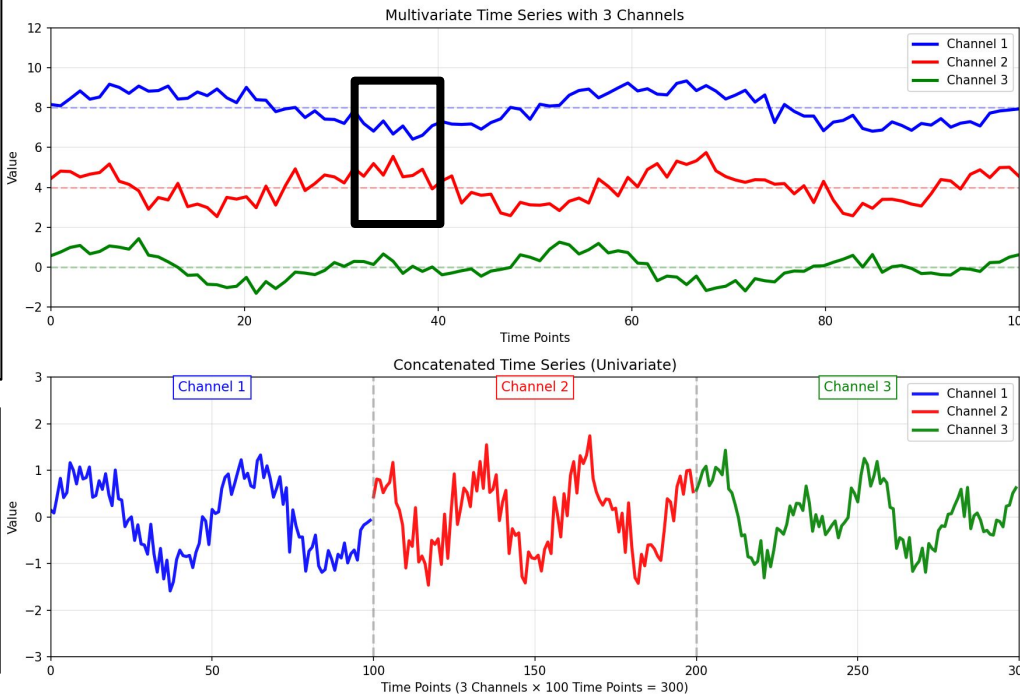
Strong dependency: Classification is difficult using any single channel alone.

Weak dependency: A single channel has sufficient discriminative information.

Foundation Model Approaches:

- Tokenization-based (Chronos)
- Patch-based (Mantis, Moment)
- Shape-based (VQShape)

All these approaches process channels independently, reducing complexity from $O((L \cdot C)^2)$ to $O(C \cdot L^2)$ but missing critical cross-channel interactions.



Evaluation Datasets

Counter Movement Jump (CMJ)

Accelerometer data (3 channels)

Weak: class distinctions primarily in y-channel

Military Press (MP)

Motion capture data (8 or 50 channels)

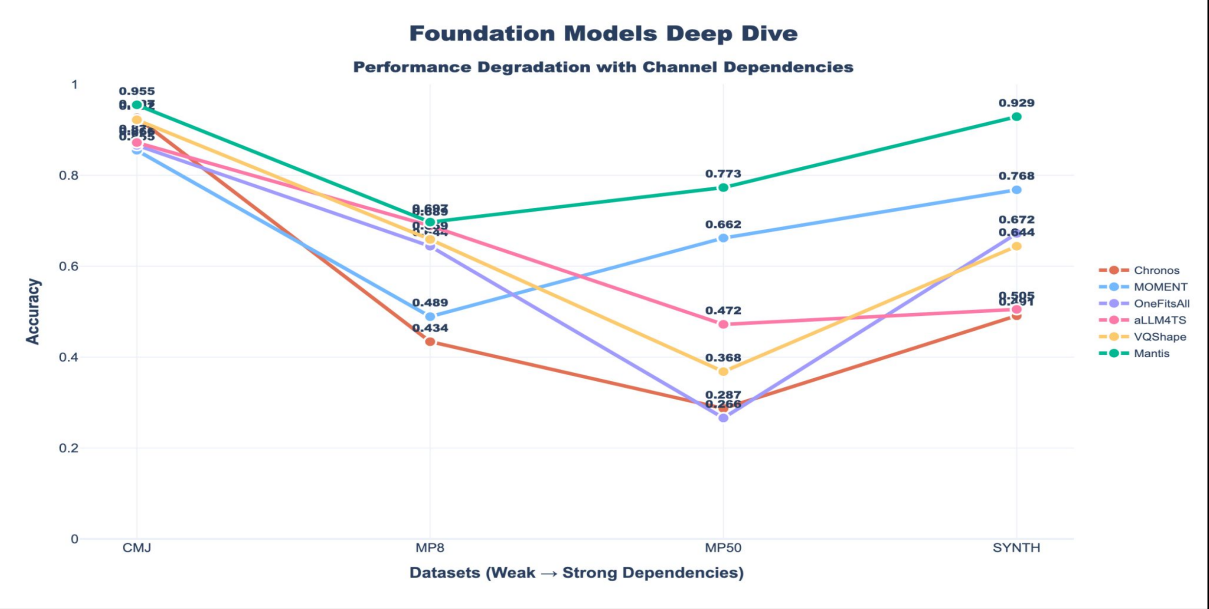
Strong: requires minimum 4 channels

Synthetic (SYNTH)

Designed dataset (8 channels)

Strong: features in only 2 random channels (requires min 2 channels)

Method Category	Method	CMJ (Weak)	MP8 (Strong)	MP50 (Strong)	SYNTH (Strong)
Time Series Specific	ROCKET	0.950	0.743	0.793	0.861
	QUANT	0.933	0.696	0.740	0.964
	MiniRocket	0.944	0.741	0.787	0.882
Deep Learning	InceptionTime	0.905	0.832	0.781	0.691
	ConvTran	0.866	0.804	0.570	0.930
Foundation Models	Mantis (best)	0.955	0.697	0.773	0.929
	MOMENT	0.855	0.489	0.662	0.768
	Chronos (best)	0.927	0.467	0.287	0.513
	aLLM4TS (best)	0.872	0.689	0.472	0.505



Why Is Mantis Better Than Other FMs?

Partially Channel-Aware Design: foundation model with channel-level adapters (PCA, SVD, Linear Combiner) - others completely ignore multivariate structure.

BUT:

Mantis

Accuracy (MP50)
77.3%

Fine-tuning Time
Hours

Model Complexity
Very High

Has channel adapters
(PCA, SVD)

MiniRocket

Accuracy (MP50)
78.7%

Training Time
4.86 secs

Model Complexity
Low

No adapters needed

Conclusions:

- Many foundation models **process channels independently**.
- Foundation models fail as data complexity increases. They **gain efficiency by ignoring channels** $O(C \cdot L^2)$ vs $O(L \cdot C)^2$ but **lose accuracy** on complex tasks.
- **Time series specific methods** consistently **outperform** foundation models on complex datasets.
- **"Bigger" doesn't mean "better"** for time series. Domain-specific approaches still lead when data relationships matter.

Future work might focus on:

- Data Representation
- Hybrid Architectures
- Channel-Aware Pre-training

Code & Data:

